

Direct and Indirect Effects: Classical and Bootstrap Estimates of Variability

Author(s): Kenneth A. Bollen and Robert Stine

Source: *Sociological Methodology*, 1990, Vol. 20 (1990), pp. 115-140

Published by: American Sociological Association

Stable URL: <https://www.jstor.org/stable/271084>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



Wiley and American Sociological Association are collaborating with JSTOR to digitize, preserve and extend access to *Sociological Methodology*

JSTOR

DIRECT AND INDIRECT EFFECTS: CLASSICAL AND BOOTSTRAP ESTIMATES OF VARIABILITY

*Kenneth A. Bollen** and
Robert Stine†

The decomposition of effects in structural equation models has been of considerable interest to social scientists. Finite-sample or asymptotic results for the sampling distribution of estimators of direct effects are widely available. Statistical inferences about indirect effects have relied exclusively on asymptotic methods which assume that the limiting distribution of the estimator is normal, with a standard error derived from the delta method. We examine bootstrap procedures as another way to generate standard errors and confidence intervals and to estimate the sampling distributions of estimators of direct and indirect effects. We illustrate the classical and the bootstrap methods with three empirical examples. We find that in a moderately large sample, the bootstrap distribution of an estimator is close to that assumed with the

The authors names are listed alphabetically to represent equal contribution. We gratefully acknowledge support from the University Research Council of the University of North Carolina. We thank Walter Davis for computational assistance and the referees for helpful comments. An earlier version of this paper was presented at the 1988 Annual Meetings of the American Sociological Association, Atlanta.

*University of North Carolina at Chapel Hill

†University of Pennsylvania

classical and delta methods but that in small samples, there are some differences. Bootstrap methods provide a check on the classical and delta methods when the latter are applied under less than ideal conditions.

1. INTRODUCTION

Sampling distributions are central to inference, and much statistical work centers on obtaining these distributions under appropriate regularity conditions. But sometimes the complexity of such derivations pushes one from the real problem to a more tractable setting, such as that provided by normality or large samples. Often such conditions are not met (e.g., when we have nonnormal data or small samples), and the distribution of our estimator is unknown. For instance, although the sampling distribution of an ordinary least squares (OLS) estimator of a regression coefficient approximates a normal distribution in large samples (Johnston 1984, pp. 279–81), the ratio of a coefficient estimator to its standard error need not follow a t distribution in small samples unless the unobserved disturbances are normally distributed.

We focus on the properties of estimators of direct and indirect effects in structural equation models through three examples that compare traditional methods to recent resampling approaches. The decomposition of effects in structural equation (path) models has been the subject of much research (Lewis-Beck 1974; Alwin and Hauser 1975; Fox 1980). One part of this effort has been devoted to testing the statistical significance of direct, indirect, and specific effects (Folmer 1981; Sobel 1982, 1986; Bollen 1987). Typically, the standard errors and tests of statistical significance for these are based on asymptotic theory. Folmer and Sobel, for instance, recommend the delta method to estimate the standard errors and statistical significance of total and indirect effects. But they note that the properties of the delta method are unknown in small samples. A further complication concerns the standardized direct and indirect effects, which require adjustments that are not always easy to apply.

Our paper illustrates that bootstrap standard errors and confidence intervals for direct and indirect effects offer an alternative. Efron is the best-known advocate of the bootstrap methodology

(Efron 1979, 1982; Efron and Gong 1983; Efron and Tibshirani 1986). He has demonstrated the utility of these methods with correlation coefficients, Cox's proportional hazard rate models, regression analysis, autoregressive time-series, and other problems. New applications for bootstrap methods appear regularly (e.g., Freedman and Peters 1984; Chatterjee 1984). In section 2, we give a brief overview of the workings of the bootstrap. In section 3, we examine direct effects, and in section 4, we examine indirect effects. We conclude with a discussion of our findings and how they might apply to other problems.

2. BOOTSTRAP OVERVIEW

The bootstrap is an approach to estimating properties of estimators based on samples drawn from the original observations. By now, several expository accounts (e.g., Efron and Tibshirani 1986; Stine 1989; Dietz, Frey, and Kalof 1987) and technical accounts (e.g., Beran 1984; Efron 1982; Freedman 1981) are available.

Suppose that $\{Z_1, Z_2, \dots, Z_N\}$ is a random sample with unknown distribution function, F , characterized by a parameter θ . Our estimator of θ is symbolized as $\hat{\theta}$, and we wish to know its population variance, $\text{VAR}(\hat{\theta})$, and its sampling distribution. The bootstrap approach relies on sampling the empirical distribution function of the data, F_N , to estimate properties of $\hat{\theta}$. For the random variables, $\{Z_1, Z_2, \dots, Z_N\}$, we observe a given sample $\{z_1, z_2, \dots, z_N\}$. The empirical distribution function, F_N , is that distribution with mass $1/N$ on $z_1, z_2, \dots, z_N$. If we take N independent draws *with replacement* from F_N , we have a bootstrap sample $Z^* = \{Z_1^*, Z_2^*, \dots, Z_N^*\}$. Using the same formula for $\hat{\theta}$, we can compute $\hat{\theta}^*$ using the bootstrap sample. Repeatedly sampling with replacement N cases from F_N and computing $\hat{\theta}^*$ leads to $\hat{\theta}^*(1), \hat{\theta}^*(2), \dots, \hat{\theta}^*(B)$, where the numbers in parentheses distinguish the bootstrap samples ($b = 1, 2, \dots, B$) and B is the total number of bootstrap samples drawn. These replications of $\hat{\theta}$ determine the bootstrap distribution of $\hat{\theta}^*$ and the bootstrap estimate of $\text{VAR}(\hat{\theta}^*)$,

$$\text{var}(\hat{\theta}^*) = \sum_{b=1}^B (\hat{\theta}^*(b) - \bar{\hat{\theta}^*})^2 / (B-1), \quad (1)$$

where $\bar{\hat{\theta}}^* = \sum_{b=1}^B (\hat{\theta}^*(b))/B$ is the average of the B bootstrap replications of $\hat{\theta}$. Similarly, $[\text{var}(\hat{\theta}^*)]^{1/2}$ gives the bootstrap standard error estimate of $\hat{\theta}$. The distribution of $\hat{\theta}^*$ is an estimator of the sampling distribution of $\hat{\theta}$. Efron (1987) recommends a B on the order of 100 for standard error estimates.

This form of bootstrapping is nonparametric in that we need not make assumptions about the particular type (e.g., normal) of distribution function for F .¹ Instead, we assume that sampling from the empirical distribution, F_N , is comparable to sampling from F and that the bootstrap sampling distribution of $\hat{\theta}^*$ is comparable to the sampling distribution of $\hat{\theta}$.² As Diaconis and Efron (1983, p. 130) state, "The bootstrap has been tried on a large number of problems . . . for which the correct answer is known. The estimate it gives is a good one for such problems, and it can be mathematically proved to work for similar problems." Furthermore, Singh (1981), Bickel and Freedman (1981), Beran (1984), and others have provided conditions under which the bootstrap distribution is highly accurate asymptotically. However, standard asymptotic methods, like the many descendants of the central limit theorem, have the same validity. So why bootstrap? First, it is often easier and faster to simulate than to perform the mathematical computations. For example, Beran and Srivastava (1985) show some bootstrap estimates that are far easier to obtain than their theoretic counterparts, but just as accurate. Second, bootstrap distributions often converge faster to the true distribution than straightforward asymptotic approximations (Singh 1981; Efron 1987).

Bootstrap resampling for regression models is a generalization of the bootstrap process described above. Rather than sample scalars, we sample a vector of values for each observation and compute the regression coefficient estimator for each bootstrap sample. Consider the regression model, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\mathbf{X}$ is an $N \times k$ matrix of the explanatory variables (including a column of ones for the constant term), $\boldsymbol{\beta}$ is a $k \times 1$ vector of population regression

¹ Parametric bootstrap procedures are discussed in Efron (1982).

² The bootstrap methodology can fail if θ depends too much on one part of the distribution of F . The statistic must respond smoothly to changes in the distribution. See Bickel and Freedman (1981) and Parr (1985) for "smoothness" conditions.

coefficients, ϵ is an $N \times 1$ vector of random disturbances with $E(\epsilon) = \mathbf{0}$ and $E(\mathbf{X}'\epsilon) = \mathbf{0}$, and y is the $N \times 1$ dependent variable.

The bootstrap estimate of the covariance matrix of $\hat{\beta}^*$ is $\text{var}(\hat{\beta}^*) = \frac{\sum_{b=1}^B (\hat{\beta}^*(b) - \bar{\hat{\beta}}^*) (\hat{\beta}^*(b) - \bar{\hat{\beta}}^*)'}{B-1}$, where the coefficient vector $\hat{\beta}^*(b)$ is the value of the bootstrap estimator in the b th sample, and $\bar{\hat{\beta}}^*$ is the mean of the bootstrap estimates for B bootstrap samples. The square root of the main diagonal of this matrix contains the bootstrap estimates of the standard deviations of the coefficient estimates. The collection of the $\hat{\beta}^*(b)$ forms the bootstrap distribution of the regression coefficient estimates (Efron 1982).

The bootstrap is also useful in forming confidence intervals. The simplest nonparametric bootstrap confidence interval is known as the percentile interval. The $(1 - \alpha)$ percentile interval for a statistic is the interval that captures the inner 100 $(1 - \alpha)$ percent of the bootstrap distribution. Suppose we have 99 bootstrap replications of $\hat{\beta}^*(b)$ and we want to use these to form a 90 percent confidence interval. Putting the replications into order, we have the 100 intervals $[-\infty, \hat{\beta}^*(1)]$, $[\hat{\beta}^*(1), \hat{\beta}^*(2)]$, $\dots$, $[\hat{\beta}^*(98), \hat{\beta}^*(99)]$, and $[\hat{\beta}^*(99), +\infty]$. The union of the inner 90 intervals is the 90 percent bootstrap percentile interval, $[\hat{\beta}^*(5), \hat{\beta}^*(95)]$. Since the percentile interval is based on order statistics of the collection of bootstrap replications, it need not be symmetric. Efron (1987) generally recommends a B of 1,000 or more for accurate confidence intervals.

A key feature of the percentile interval is that it automatically makes use of a transformation to normality, if one exists. As long as we are willing to assume that there exists some monotone transformation g so that

$$g(\hat{\theta}) - g(\theta) \sim N(0, \nu^2) \quad (2)$$

and

$$g(\hat{\theta}^*) - g(\hat{\theta}) \sim N(0, \nu^2), \quad (3)$$

where ν^2 is the variance, then the percentile interval works. Unfortunately, (2) and (3) do not always hold.³ For example, conditions

³ The bootstrap distribution of $g(\hat{\theta}^*)$ is discrete and cannot be normal; therefore, we should interpret (2) to mean that the bootstrap distribution becomes very close to normal as B grows larger.

(2) and (3) do not hold when $\hat{\theta}$ is biased. A simple adjustment exists in this case, and the modified interval is known as the bias-corrected percentile interval. Details of calculation and examples with the correlation coefficient appear in Efron (1982) and Stine (1989). We use only the percentile intervals because the bias-corrected intervals in our examples are virtually the same. Many more complex variations of bootstrap intervals exist, particularly Efron's (1987) accelerated intervals and Beran's (1987) prepivoted intervals.

Bootstrap procedures easily generalize to estimators of standardized regression coefficients, indirect effects, and total effects. For the standardized regression coefficients, we form the standardized estimate for each bootstrap sample and then analyze the bootstrap distribution of these in the same way we analyze the unstandardized coefficient. Similarly, we form the indirect or total effects in each bootstrap sample and investigate the bootstrap distribution for estimates of variance and for the creation of confidence intervals.

3. DIRECT EFFECTS

Direct effects are the influences of one variable on another that are not mediated by any other variable in a model, such as the usual regression coefficient estimates. Their standard errors are usually straightforward to obtain. For instance, in an OLS regression model under the usual assumptions, the variances of the regression coefficient estimators are in the main diagonal of $\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$, where σ^2 is the disturbance variance and $\mathbf{X}$ is the matrix of explanatory variables.⁴ Substituting an unbiased sample estimator for σ^2 and taking the square root of the diagonal elements of $\hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}$ lead to standard errors.

For maximum likelihood estimators (MLEs) of structural equations, the asymptotic variances of the estimators are in the main diagonal of the inverse of the information matrix

$$\{-E[\partial^2 \ln L(\boldsymbol{\theta})/\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}']\}^{-1}, \quad (4)$$

⁴ With random regressors, the variances are in the main diagonal of $\sigma^2 E(\mathbf{X}'\mathbf{X})^{-1}$. We estimate $E(\mathbf{X}'\mathbf{X})^{-1}$ by $(\mathbf{X}'\mathbf{X})^{-1}$.

where θ is the vector of parameters to be estimated and $\ln L(\theta)$ is the natural log of the likelihood function. We estimate this matrix by substituting the sample estimates for the unknown parameters. In general, these estimates of the variance of the MLEs hold in large samples, but the sample size required for them to be accurate is typically unknown unless special conditions like normality hold.

Frequently, the standard errors of standardized regression coefficients are estimated in the same manner as their unstandardized counterparts. The standardized regression coefficient is $\hat{\beta}_{yx}(s_x/s_y)$, where $\hat{\beta}_{yx}$ is the estimated unstandardized regression coefficient for the effect of x on y , s_x is the sample standard deviation of x , and s_y is the sample standard deviation of y . Researchers often estimate the standard error of the standardized coefficient as the square root of

$$\text{var}(s_x/s_y \hat{\beta}_{yx}) = s_x^2/s_y^2 \text{var}(\hat{\beta}_{yx}). \quad (5)$$

Or, equivalently, they take the standard errors from a regression analysis in which all variables have been standardized to a variance of 1. This strategy ignores the sampling variation in the estimated standard deviations and need not be accurate in small samples (Bentler and Lee 1983).

The bootstrap estimate of variances for the unstandardized direct effects should be close to those from the classical regression model when the assumptions of the regression model hold. In fact, one can show that as N increases, the two estimates converge to the same value (Freedman 1981). The differences in the variance estimates for standardized coefficients in small samples may be greater. To explore these issues, we turn to our first example.

3.1. *Example 1: Political Democracy in Latin America*

Our first example is a causal chain model:

$$\begin{aligned} y_1 &= \beta_{10} + \gamma_{11} x_1 + \epsilon_1, \\ y_2 &= \beta_{20} + \beta_{21} y_1 + \epsilon_2, \\ y_3 &= \beta_{30} + \beta_{32} y_2 + \epsilon_3, \end{aligned} \quad (6)$$

where we assume that $E(\epsilon_i) = 0$, $\text{COV}(x_1, \epsilon_i) = 0$, and ϵ_i is homoscedastic and nonautocorrelated for $i = 1, 2, 3$, and $\text{COV}(\epsilon_i, \epsilon_j) = 0$

for $i \neq j$. The variables are as follows: y_1 = industrialization in 1965 (ln energy consumption per capita); y_2 = percentage of the adult population that was literate in 1965; y_3 = the political democracy index for 1965; and x_1 = the year of economic "takeoff," i.e., the year a country began a sustained period of economic growth. The first equation of the model derives from the hypothesis that early economic takeoff dates are associated with higher societal industrialization. The second equation specifies that literacy is dependent on industrialization. And finally, the third equation represents literacy as a determinant of political democracy. Variables are for 20 Latin American countries. The data and sources are presented in Table 1.⁵

The equations in (6) and the accompanying assumptions lead to a fully recursive model. The OLS estimators of the regression coefficients for each equation are unbiased and consistent for all recursive models. Furthermore, if the disturbances (ϵ_i 's) are normally distributed, OLS is equivalent to the full-information MLE (Johnston 1984, p. 467–69).⁶

Our analyses revealed outlying cases for each equation.⁷ Should these outliers be left in? First, on practical grounds, we should keep them, because the removal of cases that are outliers in any equation would reduce the sample from 20 to 11! Second, these outliers may accurately reflect the kind of discrepancies we will find in analyzing similar data in other samples. Third, removal of cases does not solve the problems that may have produced the deviant observations. Our model is simple and underspecified. The outliers may reflect the omission of important explanatory variables. Finally,

⁵ The table contains rounded versions of the variables used in our analysis.

⁶ We tested the OLS residuals from each equation for normality (see Bollen 1989, pp. 418–23, for test statistic) and found that the residuals from the y_1 equation indicated nonnormality.

⁷ The outliers in the first equation are Trinidad and Tobago, Venezuela, Panama, and Jamaica; the outliers in the second equation are Trinidad and Tobago, Guatemala, Venezuela, and Bolivia; and the outliers in the third equation are Argentina, Panama, Ecuador, and the Dominican Republic. The scattergram of y_2 against y_1 suggested mild curvilinearity for the variables in the second equation. However, adding a quadratic term, double logging energy consumption per capita, or using the logit form of literacy did not lead to substantial improvements in fit.

TABLE 1
Data and Sources for Example 1

Country	Takeoff Year ^a	In Energy Consumption Per Capita in 1965	Literacy in 1965	Political Democracy in 1965
Argentina	1924	7.20	91.0	52.6
Bolivia	1958	5.22	32.0	36.2
Brazil	1946	5.85	61.0	60.9
Chile	1922	6.99	84.0	97.0
Colombia	1953	6.28	65.0	71.4
Costa Rica	1948	5.72	84.3	90.1
Dominican Republic	1958	5.27	64.5	38.8
Ecuador	1963	5.36	67.0	44.6
El Salvador	1960	5.12	49.0	72.1
Guatemala	1964	5.20	38.0	39.5
Honduras	1968	5.02	45.0	50.0
Jamaica	1962	6.79	85.0	90.1
Mexico	1934	6.88	65.0	74.5
Nicaragua	1958	5.46	49.6	55.4
Panama	1950	7.16	73.0	76.9
Paraguay	1973	4.84	68.0	44.7
Peru	1950	6.38	61.0	87.0
Trinidad and Tobago	1962	8.16	80.0	84.7
Uruguay	1938	6.82	90.3	99.6
Venezuela	1950	8.00	80.0	73.4

Sources: Collier 1975, p. 341 (takeoff year); Taylor and Hudson 1971, 1972 (energy consumption and literacy); and Bollen 1980 (political democracy).

^aFor Jamaica and Trinidad and Tobago, the takeoff year is assumed to be the year of independence.

including these outliers allows us to illustrate their impact on the bootstrap distribution and standard errors.⁸

In the equations below, we list the OLS estimates (and standard errors) on the left and the mean of the bootstrap estimates

⁸ Robust regression estimation could be used to lessen the potential effects of the outliers on the parameter estimates. These robust procedures could be bootstrapped as well (see Dietz, Frey, and Kalof 1987).

(and standard deviations) for the bootstrap distribution with 1,000 replications ($B = 1,000$) on the right:⁹

$$\begin{array}{ll} \hat{y}_1 = 83.4 - 0.04x_1, & \hat{y}_1 = 85.6 - 0.04x_1, \\ (28.7) \quad (0.01) & (22.5) \quad (0.01) \end{array}$$

$$\begin{array}{ll} \hat{y}_2 = -5.8 + 11.7y_1, & \hat{y}_2 = -7.3 + 12.0y_1, \\ (18.1) \quad (2.9) & (19.0) \quad (2.9) \end{array} \quad (7)$$

$$\begin{array}{ll} \hat{y}_3 = 15.1 + 0.78y_2, & \hat{y}_3 = 14.8 + 0.78y_2. \\ (14.8) \quad (0.22) & (12.6) \quad (0.20) \end{array}$$

Generally, the bootstrap standard errors are close to the OLS standard errors.

In Figure 1, the ratio of $\hat{\beta}_{32}^*$ to its standard error is superimposed on a t distribution with $N-2$ ($=18$) degrees of freedom.¹⁰ The bootstrap distribution is slightly asymmetric and has a broad, almost bimodal peak. This unusual feature is due to the outliers in the third equation. As a result, the bootstrap intervals differ slightly from the regular intervals.

Table 2 gives the 90 percent confidence intervals for the unstandardized coefficients in the political democracy example. The classical confidence interval for, say, $\hat{\beta}_{32}$ is $\hat{\beta}_{32} \pm t_{\alpha/2}S(\hat{\beta}_{32})$, $[0.406, 1.15]$, where S equals the standard error. The estimated bootstrap percentile confidence interval is $[0.429, 1.06]$. The classical confidence interval has a range of 0.745, about 20 percent larger than the range of the bootstrap interval. Similarly, the classical confidence intervals for $\hat{\gamma}_{11}$ and $\hat{\beta}_{21}$ are wider than the bootstrap confidence intervals.

Classical confidence intervals are constructed under the assumption that the distribution of the coefficient estimator is symmetric, i.e., that the upper and lower bounds are the same distance

⁹ These data are not a random sample from a population in the usual sense. This raises the common problem of interpreting significance tests when the data consist of all or nearly all cases in a population. Both the bootstrap and the standard tests of significance address this ambiguity. See Blalock (1972, pp. 238–39), Finch (1979), and Freedman and Lane (1983) for further details on significance testing under these conditions.

¹⁰ In this and the other figures, we used “smoothed” histograms to eliminate much of the random noise. See Silverman (1986) for details.

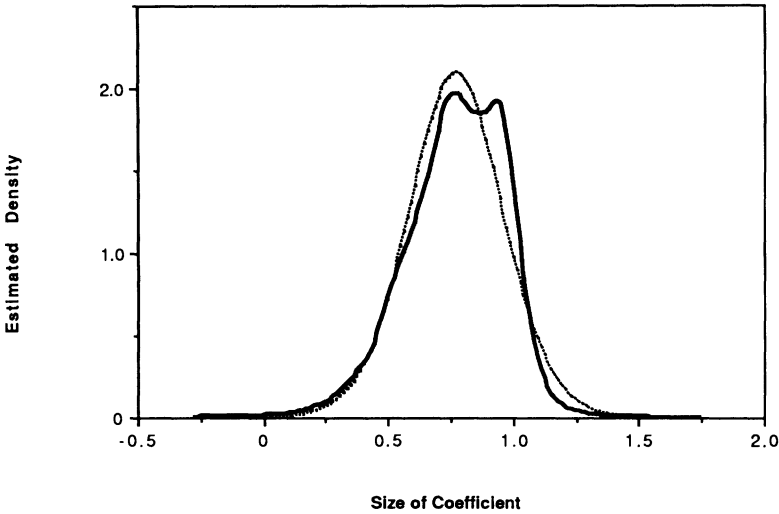


FIGURE 1 Bootstrap distribution of $\hat{\beta}_{32}^*$ from example 1 superimposed on a t density with the same mean and variance and 18 degrees of freedom.

TABLE 2
Ninety Percent Confidence Intervals for Unstandardized Direct Effects in Recursive Model of Political Democracy ($N=20$, $B=1,000$)

Confidence Intervals	Unstandardized Coefficient (Estimate)		
	$\hat{\gamma}_{11}$ ($=-0.040$)	$\hat{\beta}_{21}$ ($=11.71$)	$\hat{\beta}_{32}$ ($=0.778$)
Classical	-0.065 to -0.014	6.70 to 16.73	0.406 to 1.15
Bootstrap percentile	-0.058 to -0.021	7.63 to 17.17	0.429 to 1.06

from the coefficient estimate. The bootstrap confidence interval can capture asymmetries in the distribution of the estimator so that the lower bound can be further or closer to the coefficient estimate than the upper bound. For instance, the classical confidence interval for $\hat{\beta}_{32}$ is $(0.778 - 0.373)$ to $(0.778 + 0.373)$, and the bootstrap percentile interval is $(0.778 - 0.349)$ to $(0.778 + 0.282)$.

What happens with the standardized regression coefficients? Table 3 gives the standard errors for the three standardized

TABLE 3
Three Estimates of Standard Errors for Standardized Regression Coefficient
Estimates for Recursive Model of Political Democracy ($N = 20$)

Standardized Coefficient	Standardized OLS SE	Normal Theory SE	Bootstrap SE ($B=1,000$)
$\hat{\gamma}_{11}$	0.199	0.173	0.178
$\hat{\beta}_{21}$	0.170	0.127	0.087
$\hat{\beta}_{32}$	0.179	0.140	0.151

regression coefficients from the usual method of multiplying the OLS standard error by the ratio of the standard deviations, the normal theory standard error for correlation coefficients (Johnson and Kotz 1970, p. 229), and the bootstrap standard error. The normal theory standard error for correlations applies because the standardized regression coefficients in these equations are Pearson's correlation coefficients.

For each coefficient, the standardized OLS standard errors are consistently higher than the other standard errors. This shows that the conventional practice can overestimate the variance of the standardized coefficient. For two of the three coefficients, the normal theory standard errors are less than the bootstrap standard errors, which is consistent with Efron's analysis of a correlational example with an N of 15 (Efron and Gong 1983). However, the second coefficient estimate demonstrates that the normal theory standard error need not always be lower than the bootstrap standard error. See Efron and Gong (1983) or Efron and Tibshirani (1986) for further discussion.

In sum, in this example the bootstrap standard errors for the unstandardized coefficients are close to the OLS standard errors, even in a very small sample. The distributions of the bootstrap estimates are only slightly nonnormal; therefore, the bootstrap confidence intervals are similar to the classical confidence intervals. For the standardized coefficients, the standardized OLS standard errors consistently overestimate the standard errors, compared with the normal theory and bootstrap standard errors.

3.2. *Example 2: Marginal Propensity to Consume*

This example is a two-equation nonrecursive model suggested by Haavelmo (1953):

$$y_1 = y_2 + x_1, \quad (8)$$

$$y_2 = \beta_{21} y_1 + \epsilon_2, \quad (9)$$

where y_1 = disposable income, y_2 = consumption expenditures, and x_1 = investment expenditures. It is assumed that $\text{COV}(x_1, \epsilon_2) = 0$, that $E(\epsilon_2) = 0$, and that ϵ_2 is homoscedastic and nonautocorrelated. The series are U.S. annual figures from 1922 to 1941. The data are listed in Haavelmo (1953) and in Bollen (1989, p. 25). Unlike our first example, this is nonrecursive and the OLS estimator is inconsistent if applied to equation (9). We use an instrumental variable (IV) estimator instead:

$$\hat{\beta}_{21} = 1 - \text{var}(x_1) / [\text{cov}(x_1, y_1)]^2, \quad (10)$$

which is consistent and asymptotically normal with an asymptotic variance of

$$\text{AVAR}(\hat{\beta}_{21}) = [\text{VAR}(\epsilon_2)/N] [\text{VAR}(x_1)] / [\text{COV}(x_1, y_1)]^2. \quad (11)$$

We substitute sample values for the population quantities to estimate the asymptotic variance. We obtain a $\hat{\beta}_{21}$ of 0.672 with an estimated asymptotic standard error of 0.035 and a t ratio of 19.2. To associate a p value with this critical ratio or to provide a confidence interval, the sample must be sufficiently large for the estimated asymptotic standard error to be accurate and for the distribution of the estimator to be normal. The small sample size is worrisome, particularly since the bootstrap standard error is 0.046 ($B = 1,000$), which is about 30 percent larger than the estimated asymptotic standard error.

Is the distribution of $\hat{\beta}_{21}$ approximately normal? In Figure 2, the bootstrap distribution of $\hat{\beta}_{21}^*$ is superimposed on a normal distribution with the same standard deviation. The bootstrap distribution exhibits more of a peak than the normal distribution and is asymmetrical. Also noteworthy is the “bump” on the right side of the bootstrap distribution. Again, this unusual feature is due to an outlier in the last year in the data set (see Bollen 1989, pp. 25–28). The outlier suppresses $\hat{\beta}_{21}$, which is 0.715 if the last year is removed.

Can the single outlier account for differences between the bootstrap and the asymptotic standard errors? Rerunning the 1,000 bootstrap samples without the last year leads to an estimated asymptotic standard error of 0.038 and a bootstrap standard error of 0.071, which is over 85 percent higher. Is the outlier responsible for the nonnormal distribution of $\hat{\beta}_{21}^*$? The bootstrap distribution

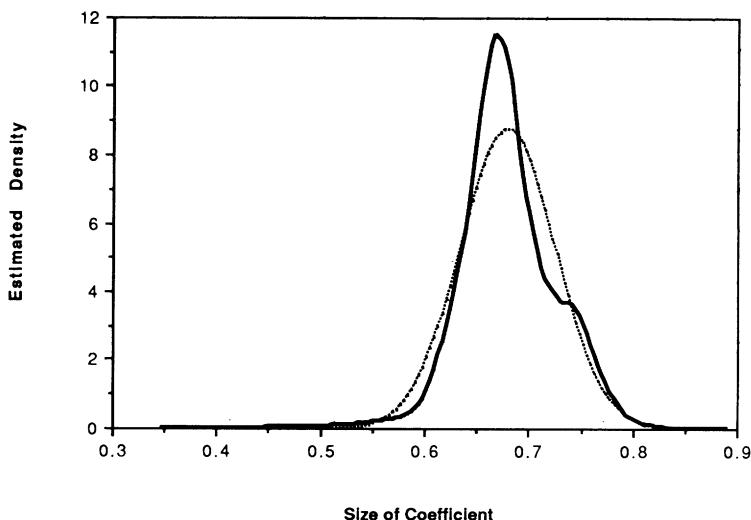


FIGURE 2 Bootstrap distribution of $\hat{\beta}_{21}^*$ from example 2 superimposed on a normal density with the same mean and variance.

without the outlier is not as peaked as it is with it, but it is still asymmetric. Figure 3 is a Q-Q normal plot that compares the bootstrap distribution with a normal distribution. If the bootstrap distribution of $\hat{\beta}_{21}^*$ were normal, it would fall on top of the straight line in the body of the plot. Obviously, the distribution of $\hat{\beta}_{21}^*$ is nonnormal.

As might be expected, the usual confidence intervals differ from the bootstrap confidence intervals. For the sample in which the last year is removed, the 90 percent confidence intervals are [0.649, 0.781] for the IV estimator with the estimated asymptotic standard error and [0.602, 0.768] for the bootstrap percentile interval. The lower bound (and to a lesser extent the upper bound) of the IV 90 percent confidence interval is higher than the bootstrap confidence intervals. The bootstrap intervals also capture the asymmetry of the distribution of $\hat{\beta}_{21}^*$ with a wider interval below 0.715 than above it. In contrast, the IV confidence interval is symmetric. Generally, for this example we find that the bootstrap estimates of variability differ considerably from those predicted from asymptotic theory and that the distribution of the bootstrap estimates is nonnormal.¹¹

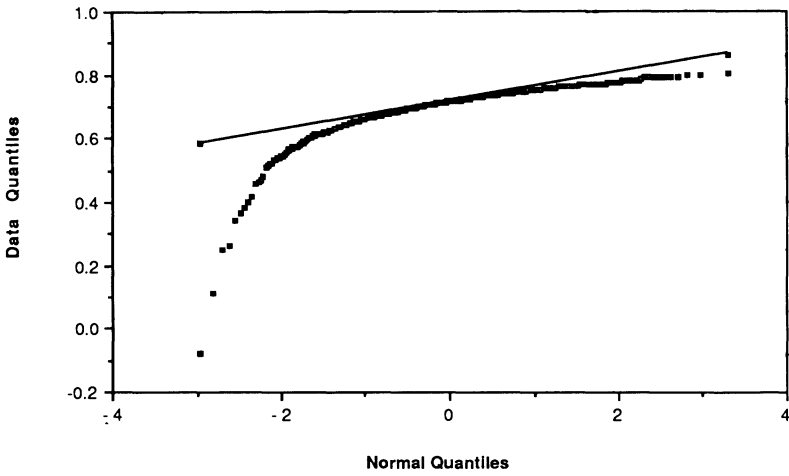


FIGURE 3 QQ plot of $\hat{\beta}_{21}^*$ from example 2 without the outlier.

4. INDIRECT EFFECTS

An indirect effect is the effect of one variable on another that is mediated by at least one other variable in a model. In example 1 (section 3.1), the economic takeoff year (x_1) has an indirect effect on political democracy (y_3) through x_1 's effect on industrialization (y_1), industrialization's effect on literacy (y_2), and literacy's direct effect on political democracy. This estimated indirect effect is $\hat{\gamma}_{11}\hat{\beta}_{21}\hat{\beta}_{32}$, and we need a standard error and knowledge of the distribution of the product to test its significance.

Folmer (1981, pp. 1440–42) and Sobel (1982, 1986) suggested that the delta method could be applied to estimate the asymptotic variances of indirect (and total) effects. The delta method is a method of estimating the variance of a nonlinear combination of parameter estimators. Indirect effects are an example of a nonlinear function of several coefficient estimators. The basic idea of the delta

¹¹ We also found a potential autocorrelation problem with the disturbances. For the sample with 19 observations, the autocorrelation was 0.39, which is of borderline statistical significance. More complicated bootstrap methods could be devised to further investigate this issue. See Efron and Tibshirani (1986) for applications of the bootstrap to time-series data.

method is to approximate a nonlinear function by a linear function of the form

$$f(\hat{\theta}) \approx f(\theta) + (df(\theta)/d\theta) (\hat{\theta} - \theta). \quad (12)$$

Expression (12) leads to an approximate variance of

$$\text{VAR}(f(\hat{\theta})) \approx (df(\theta)/d\theta)^2 \text{VAR}(\hat{\theta}). \quad (13)$$

The accuracy of the delta method variance approximation depends on the distance of $\hat{\theta}$ from θ , the smoothness of the derivative, the accuracy of the linear approximation (12), and the size of the sample.

To perform significance tests on $f(\hat{\theta})$, we also need to know its sampling distribution. Equation (12) shows that $f(\hat{\theta})$ is approximately equal to a linear function of $\hat{\theta}$. In large samples, $\hat{\theta}$ has a normal distribution. Since $f(\hat{\theta})$ approximately equals a constant plus a multiple of a normal variable, it approximately has a normal distribution in large samples. For further discussion of the delta method, see Rao (1973, pp. 385–89) and Bishop, Fienberg, and Holland (1975, pp. 486–500).

The indirect effect of the takeoff year (x_1) on political democracy (y_3) is $\hat{\gamma}_{11}\hat{\beta}_{21}\hat{\beta}_{32} = -0.361$. Using the delta method, we see that the estimate of the asymptotic standard error of this indirect effect is

$$\begin{aligned} & [\hat{\beta}_{21}^2\hat{\beta}_{32}^2\text{var}(\hat{\gamma}_{11}) + \hat{\gamma}_{11}^2\hat{\beta}_{32}^2\text{var}(\hat{\beta}_{21}) \\ & + \hat{\gamma}_{11}^2\hat{\beta}_{21}^2\text{var}(\hat{\beta}_{32})]^{\frac{1}{2}} = 0.189. \end{aligned} \quad (14)$$

The standard error based on the bootstrap distribution of this indirect effect is 0.166, which is about 10 percent smaller. To form tests of statistical significance with the delta method, we assume that the sampling distribution of the indirect effect estimator is approximately normal. Figure 4 shows that the normal distribution is not a good match to the bootstrap distribution, the latter being somewhat skewed.

The bootstrap confidence interval reflects the lack of symmetry in the bootstrap distribution. The 90 percent confidence intervals for the indirect effect are $[-0.672, -0.050]$ for the delta method and $[-0.653, -0.128]$ for the bootstrap percentile method. It is also interesting to see that the delta method comes close to

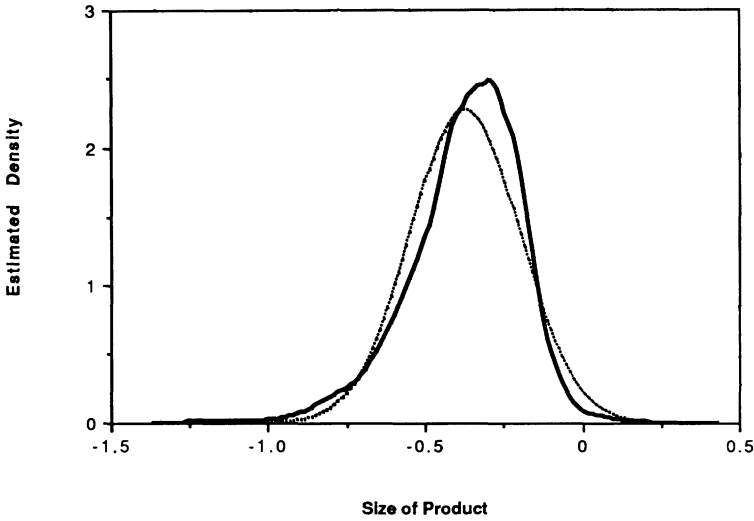


FIGURE 4 Bootstrap distribution of $\hat{\gamma}_{11}^* \hat{\beta}_{21}^* \hat{\beta}_{32}^*$ from example 1 superimposed on a normal density with matching mean and variance.

including 0 within its interval, whereas the two bootstrap confidence intervals are further from 0. In fact, for a 95 percent confidence interval, the upper bounds are +0.01 for the delta method and near -0.1 for the bootstrap method. Thus, we could be led to different statistical inferences about the indirect effect.

In example 2, we found that the asymptotic and bootstrap results for the direct effects differed. Here, our interest is in the indirect effects. The estimated indirect effect of y_1 on y_2 is $(1 - \beta_{21})^{-1} \beta_{21}^2$. For the full sample ($N=20$), the estimated indirect effect is 1.377. In Figure 5a, the bootstrap distribution of the indirect effect is superimposed on a normal distribution with the delta method standard error. A radical difference is evident. Much of this discrepancy is due to the outlier we identified earlier. When the last year is removed, the estimated indirect effect is 1.794. The estimated asymptotic standard error with the delta method is 0.435. The bootstrap standard error estimate is 0.510, which is nearly 20 percent larger. In Figure 5b, the bootstrap distribution with the outlier removed is superimposed on a normal distribution with the delta method standard error. The distribution of this indirect effect

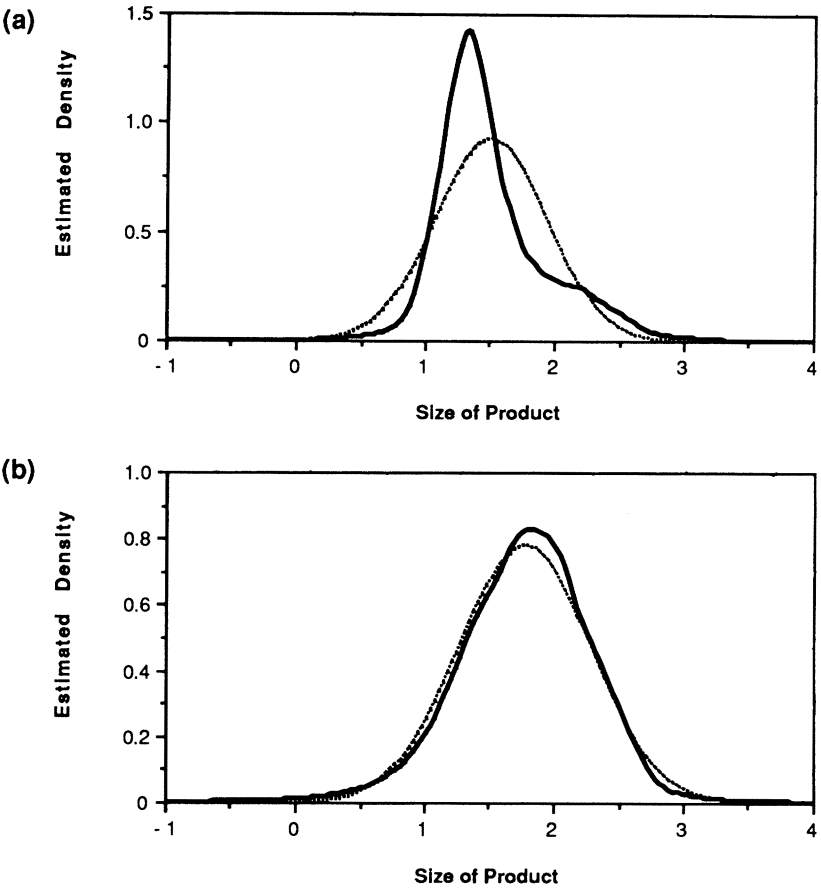


FIGURE 5 Bootstrap distribution of $(1 - \hat{\beta}_{21}^*)^{-1}(\hat{\beta}_{21}^*)^2$ from example 2 superimposed on a normal density with matching mean and variance: (a) with all 20 cases and (b) with the outlier removed.

is less extreme but still does not match a normal distribution, which is assumed when the estimate of the standard error from the delta method is used to perform significance testing.

As a result, the confidence intervals with the outlier omitted are more similar, but the bootstrap confidence intervals remain asymmetric.¹² The 90 percent confidence intervals for the indirect effect ($N = 19$) are [1.078, 2.510] for the delta method and [0.913, 2.547] for the bootstrap percentile method.

In sum, for the second example we find that for the indirect effect of y_1 on y_2 , the delta method standard error differs from the

bootstrap standard error and that the distribution of the bootstrap estimates are skewed, contrary to the normal distribution commonly assumed when the delta method standard errors are used to perform tests of statistical significance.

Lest we leave the impression that the standard errors generated by the delta method and the accompanying assumption of a normal distribution for the indirect effect estimator are always doubtful, we remind the reader that the delta method is based on asymptotic theory. That is, the results assume that a sufficiently large sample is used. Clearly, our two examples, which have around 20 cases, are small. Our next example illustrates the operation of the delta and bootstrap methods in a larger sample.

4.1. *Example 3: Union Sentiment Model*

The third example is a three-equation recursive model. The data are from McDonald and Clelland's (1984) study of union sentiment among textile workers in a southern community without any unionized mills. The model differs from McDonald and Clelland's (1984) and is taken from Bollen (1989, pp. 82–83). Figure 6 is the path diagram for the model. The variables are as follows: x_1 = years in the textile mill, x_2 = age, y_1 = deference (or submissiveness) to managers, y_2 = support for labor activism, and y_3 = sentiment toward unions.

As in example 1, if the disturbances are normally distributed, OLS applied to each equation is equivalent to the MLE. The covariance matrix for this model is

$$\mathbf{S} = \begin{bmatrix} 14.610 & -5.250 & -8.057 & -18.857 & -0.482 \\ -5.250 & 11.017 & 11.087 & 17.861 & 0.677 \\ -8.057 & 11.087 & 31.971 & 28.250 & 1.559 \\ -18.857 & 17.861 & 28.250 & 215.662 & 7.139 \\ -0.482 & 0.677 & 1.559 & 7.139 & 1.021 \end{bmatrix}.$$

¹² The indirect effects of y_1 on y_1 , of y_2 on y_1 , and of y_2 on y_2 are equivalent to the reduced-form coefficient, $\beta_{21}(1 - \beta_{21})^{-1}$, of regressing y_2 on x_1 . Under the usual assumptions for the model, OLS is an unbiased and consistent estimator of the reduced-form coefficients, and the standard errors can be obtained without using the delta method. We do not report the results here.

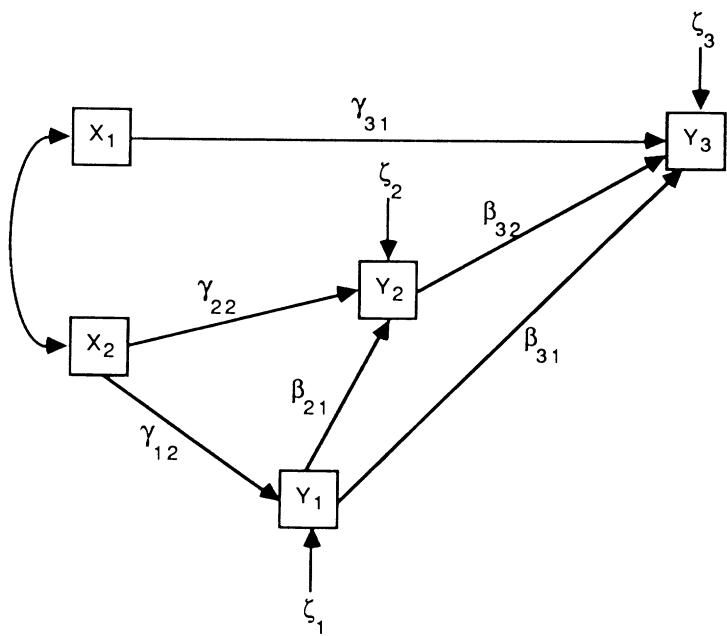


FIGURE 6 Path diagram for example 3.

The OLS estimates (and standard errors) are presented in Table 4 ($N = 173$). We found that the standard errors and confidence intervals based on the OLS results were close to those from the bootstrap.¹³ Here, our focus is on the indirect effects. Table 5 compares the bootstrap and delta method estimates of the standard errors for

TABLE 4
OLS Estimates and Standard Errors for Union Sentiment Model ($N = 173$)

	Parameter ^a					
	γ_{12}	γ_{22}	γ_{31}	β_{21}	β_{31}	β_{32}
Estimate	-0.087	0.058	0.860	-0.285	-0.218	0.850
Standard error	(0.019)	(0.016)	(0.340)	(0.062)	(0.097)	(0.112)

^aSee path diagram for paths to which parameters refer.

¹³ See Bollen and Stine (1988) for more details on this example.

TABLE 5
Bootstrap and Delta Method Estimates of Standard Errors of Indirect Paths
in the Union Sentiment Model ($N = 173$, $B = 1,000$)

Path	Delta		Bootstrap	
	Estimate	SE	Average	SE
$i_1 = \gamma_{12} \beta_{31}$	0.019	0.0094	0.019	0.0090
$i_2 = \gamma_{12} \beta_{21} \beta_{32}$	0.021	0.0070	0.021	0.0070
$i_3 = \gamma_{22} \beta_{32}$	0.049	0.0151	0.049	0.0149
$i_4 = i_3 - (i_1 + i_2)$	0.009	0.0208	0.009	0.0196

several indirect effects. The standard errors are very close for both methods. In Figure 7a, the bootstrap distribution for $\hat{\gamma}_{12}^* \hat{\beta}_{21}^* \hat{\beta}_{32}^*$ is superimposed on a normal distribution. Though not identical, these distributions are similar. The same is true for the other indirect effects. So we see that for this example, the bootstrap and delta methods give comparable results. Here, it appears that the sample of 173 is sufficiently large for the delta method and a normal distribution to work well. Indeed, if we randomly sample only 50 out of the original 173 cases and calculate the bootstrap distribution plotted against a normal distribution, we get a much more skewed distribution for the bootstrap estimates, as shown in Figure 7b.

5. CONCLUSION

Bootstrap techniques provide alternative estimates of variances and distributions for unstandardized or standardized estimators of direct and indirect effects. Perhaps our most consistent finding is the asymmetric distribution of the bootstrap estimates when the sample is small. Given the widespread use of the symmetrical normal and t distributions in the classical tests, there is a potential for incorrect inferences when the assumptions for the classical tests are not met. The bootstrap confidence intervals reflected this asymmetry, but the classical confidence intervals did not. The asymmetry was less of a problem in our example with 173 cases.

A second characteristic of two of our examples was the impact of outliers. We found that such cases introduced irregular features such as broad peaks or heavy tails into the bootstrap distributions

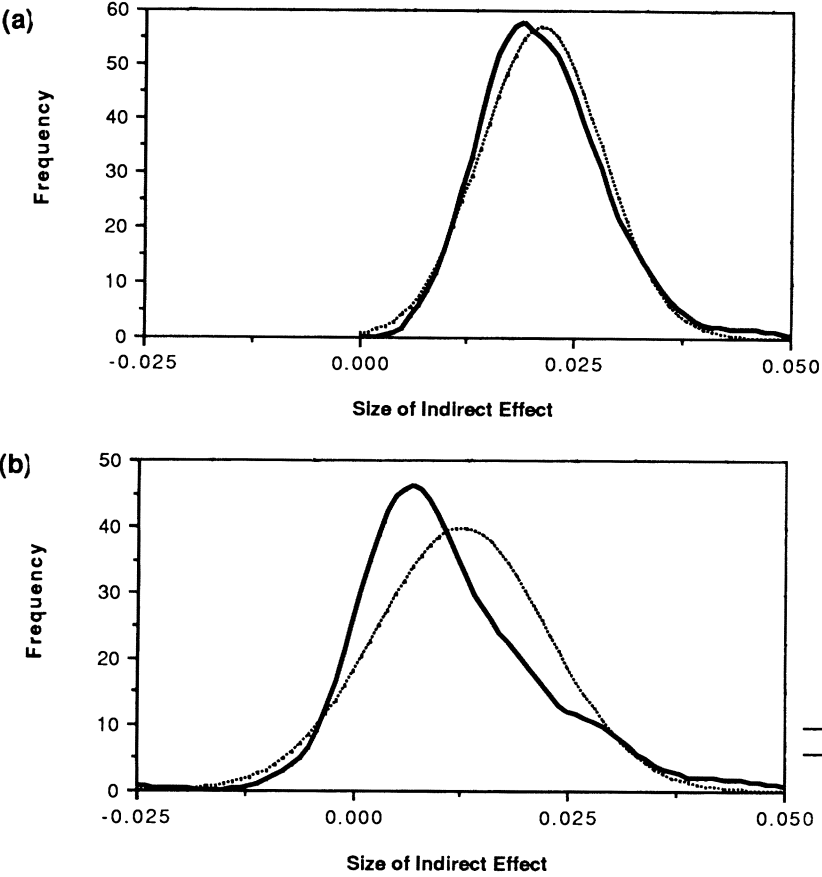


FIGURE 7 Bootstrap distribution of $\hat{\gamma}_{12}^* \hat{\beta}_{21}^* \hat{\beta}_{32}^*$ from example 3 superimposed on a normal density with matching mean and variance: (a) with $N = 173$ observations and (b) with $N = 50$ observations.

of estimates, which could affect our statistical inferences. The bootstrap methods do not remove the impact of outliers, but they show how such observations can affect the distribution of an estimator.

The standard errors from OLS for the unstandardized direct effect estimates in examples 1 and 3 were close to the bootstrap standard errors. In contrast, the estimated asymptotic standard errors for the IV estimator of direct effects in example 2 were far lower than the bootstrap standard errors. Results from example 1 showed that the standardized OLS standard errors for standardized coefficient estimates can be inaccurate. The common practice of

assuming that standardized regression coefficients can be tested with a simple adjustment of the OLS standard errors for the unstandardized estimator can lead to trouble, particularly in small samples.

For the indirect effects, we found that the bootstrap standard errors were lower than the delta method standard errors for example 1 but that the reverse was true for example 2. Again, both bootstrap distributions were asymmetric. In example 1, we saw how the choice of method could lead to different inferences about the indirect effect. Part of the problem is that the delta method is based on a large-sample approximation. For example, the Taylor series approximation to the nonlinear indirect effects uses only first-order terms. In small samples the neglected higher-order terms may have a considerable impact, which is not present with a big N . In our larger sample in example 3, the bootstrap distribution of the indirect effect estimates was close to that assumed when the delta method is used for tests of statistical significance.

Not surprisingly, our results are consistent with the suggestion that the bigger the sample, the better the classical and delta methods work. But how large is “sufficiently large”? What should be done when only a small or moderate-size sample is available? As we have demonstrated, the bootstrap provides an alternative way of studying estimators that can be compared with these classical methods. For more general structural equation models that involve iterative estimation (e.g., the ML or GLS fitting functions in LISREL or EQS), the bootstrap procedure may be too computationally demanding to run without some modifications. One possibility is to use the original model estimates as the starting values for each bootstrap sample and to take only a single step in the iterative procedure from this starting point. The resulting estimate is taken as the bootstrap estimate for that bootstrap sample, and the same is done for all bootstrap samples. Some ideas along this line appear in Jorgensen (1987).

We end with two cautionary points. First, although the bootstrap does not assume a particular distribution for the variables, it does assume that the variances and sampling distribution of the estimates from the bootstrap samples mirror those of the population distribution, F . Second, a big advantage of the bootstrap over the usual Monte Carlo simulation is that it is customized to the specific data that is being analyzed. But it shares with Monte Carlo simula-

tions the limitation that it is not always clear whether the results found in a few models hold more generally.

REFERENCES

- Alwin, D. F., and R. M. Hauser. 1975. "The Decomposition of Effects in Path Analysis." *American Sociological Review* 40:37–47.
- Bentler, P. M., and S. Y. Lee. 1983. "Covariance Structures Under Polynomial Constraints: Applications to Correlation and Alpha-Type Structural Models." *Journal of Educational Statistics* 8:207–22.
- Beran, R. 1984. "Jackknife Approximations to Bootstrap Estimates." *Annals of Statistics* 12:101–18.
- . 1987. "Prepivoting to Reduce Level Error of Confidence Sets." *Biometrika* 74:457–68.
- Beran, R., and M. S. Srivastava. 1985. "Bootstrap Tests and Confidence Intervals for Functions of a Covariance Matrix." *Annals of Statistics* 13:95–115.
- Bickel, P., and D. A. Freedman. 1981. "Some Asymptotic Theory for the Bootstrap." *Annals of Statistics* 9:1196–1217.
- Bishop, Y. M. M., S. E. Fienberg, and P. W. Holland. 1975. *Discrete Multivariate Analysis*. Cambridge, MA: MIT Press.
- Blalock, H. M. 1972. *Social Statistics*. 2d ed. New York: McGraw-Hill.
- Bollen, K. A. 1980. "Issues in the Comparative Measurement of Political Democracy." *American Sociological Review* 45:370–90.
- . 1987. "Total, Direct, and Indirect Effects in Structural Equation Models." Pp. 37–69 in *Sociological Methodology 1987*, edited by C. C. Clogg. Washington, DC: American Sociological Association.
- . 1989. *Structural Equations with Latent Variables*. New York: Wiley.
- Bollen, K. A., and R. Stine. 1988. "Bootstrapping Structural Equation Models: Variability of Indirect Effects and Goodness of Fit Measures." Paper presented at the Annual Meetings of the American Sociological Association, Atlanta.
- Chatterjee, S. 1984. "Variance Estimation in Factor Analysis: An Application of the Bootstrap." *British Journal of Mathematical and Statistical Psychology* 37:252–62.
- Collier, D. 1975. "Timing of Economic Growth and Regime Characteristics in Latin America." *Comparative Politics* 7:331–59.
- Diaconis, P., and B. Efron. 1983. "Computer Intensive Methods in Statistics." *Scientific American* 248:116–30.
- Dietz, T., R. S. Frey, and L. Kalof. 1987. "Estimation with Cross-National Data: Robust and Nonparametric Methods." *American Sociological Review* 52:380–90.
- Efron, B. 1979. "Computers and the Theory of Statistics: Thinking the Unthinkable." *Siam Review* 21:460–80.

- . 1982. "The Jackknife, the Bootstrap, and Other Resampling Plans." CBMS 38, SIAM-NSF.
- . 1987. "Better Bootstrap Confidence Intervals." *Journal of the American Statistical Association* 82:171–85.
- Efron, B., and Gail Gong. 1983. "A Leisurely Look at the Bootstrap, the Jackknife, and Cross-Validation." *American Statistician* 37:36–48.
- Efron, B., and R. Tibshirani. 1986. "Bootstrap Methods for Standard Errors, Confidence Intervals, and Other Measures of Statistical Accuracy." *Statistical Science* 1:54–74.
- Finch, P. D. 1979. "Description and Analog in Statistics." *Biometrika* 66:195–208.
- Folmer, H. 1981. "Measurement of the Effects of Regional Policy Instruments by Means of Linear Structural Equation Models and Panel Data." *Environment and Planning A* 13:1435–48.
- Fox, J. 1980. "Effect Analysis in Structural Equation Models." *Sociological Methods and Research* 9:3–28.
- Freedman, D. A. 1981. "Bootstrapping Regression Models." *Annals of Statistics* 9: 1218–28.
- Freedman, D. A., and S. Peters. 1984. "Bootstrapping a Regression Equation: Some Empirical Results." *Journal of the American Statistical Association* 79:97–106.
- Freedman, D. A., and D. Lane. 1983. "Significance Testing in a Nonstochastic Setting." Pp. 185–208 in *A Festschrift for Erich L. Lehmann*, edited by P. J. Bickel, K. A. Doksum, and J. L. Hodges, Jr. Belmont, CA: Wadsworth.
- Haavelmo, T. 1953. "Methods of Measuring the Marginal Propensity to Consume." Pp. 75–91 in *Studies in Econometric Methods*, edited by W. C. Hoods and T. C. Koopmans. New York: Wiley.
- Johnson, N., and S. Kotz. 1970. *Continuous Univariate Distributions*. Vol. 2. Boston: Houghton-Mifflin.
- Johnston, J. 1984. *Econometric Methods*. New York: McGraw-Hill.
- Jorgensen, M. A. 1987. "Jackknifing Fixed Points of Iterations." *Biometrika* 74:207–11.
- Lewis-Beck, M. S. 1974. "Determining the Importance of an Independent Variable." *Social Science Research* 3:95–107.
- McDonald, J. A., and D. A. Clelland. 1984. "Textile Workers and Union Sentiment." *Social Forces* 63:502–21.
- Parr, W. C. 1985. "The Bootstrap: Some Large Sample Theory and Connections with Robustness." *Statistics and Probability Letters* 3:97–100.
- Rao, C. R. 1973. *Linear Statistical Inference and its Applications*. 2d ed. New York: Wiley.
- Silverman, B. W. 1986. *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall.
- Singh, K. 1981. "On the Asymptotic Accuracy of Efron's Bootstrap." *Annals of Statistics* 9:1187–95.
- Sobel, M. E. 1982. "Asymptotic Confidence Intervals for Indirect Effects in Structural Equation Models." Pp. 290–312 in *Sociological Methodology 1982*, edited by S. Leinhardt. San Francisco: Jossey-Bass.

- . 1986. "Some New Results on Indirect Effects and Their Standard Errors in Covariance Structure Models." Pp. 159–86 in *Sociological Methodology 1986*, edited by N. B. Tuma. Washington, DC: American Sociological Association.
- Stine, R. 1989. "An Introduction to Bootstrap Methods: Examples and Ideas." *Sociological Methods and Research* 18: 243–91.
- Taylor, C. L., and M. C. Hudson. 1971. *World Handbook of Political and Social Indicators II*. Ann Arbor: ICPSR.
- . 1972. *World Handbook of Political and Social Indicators*. 2d ed. New Haven: Yale University Press.